

Building Technology

Basic Mathematics I for Building Technology

Professional Learner Notes

Arithmetic, fractions, decimals, percentages, ratio and measurement for building-site calculations.

Section	Purpose
Number operations	Whole numbers, decimals and calculator discipline.
Fractions and ratios	Mortar proportions, mix quantities and sharing.
Percentages	Wastage, discount, variation and increase/decrease.
Measurement	Length, area, volume and unit conversion.
Building applications	Material checks and site arithmetic.

How to Study This Topic

The learning method follows a programmed engineering mathematics approach: understand the concept, study the formula, follow a worked method, then practise with a technical problem. The aim is not memorisation only; the learner must interpret the answer in the department context.

- Write the formula before substitution.
- Use correct units.
- Check whether the result is reasonable.
- State a technical interpretation.
- Repeat practice until the method is fluent.

Formula Summary

Fraction addition

$$\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Percentage

$$\mathrm{Percentage} = \frac{\mathrm{Part}}{\mathrm{Whole}} \times 100\%$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Ratio

$$\mathrm{Ratio} = a:b$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Perimeter

$$P = 2(l+w)$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Area

$$A = lw$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

1. Number operations

Whole numbers, decimals and calculator discipline.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$$

Worked method

Example: Mortar ratio. A 1:4 mortar mix requires 50 kg cement. Estimate sand required.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports mortar mixing. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving floor areas. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving wall lengths. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving material ordering. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving mortar mixing. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving wastage allowance. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving floor areas. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving wall lengths. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving material ordering. Show formula, substitution, units and interpretation.

2. Fractions and ratios

Mortar proportions, mix quantities and sharing.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\mathrm{Percentage} = \frac{\mathrm{Part}}{\mathrm{Whole}} \times 100\%$$

Worked method

Example: Wastage. Add 8% wastage to 120 m² of floor tiles.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports wastage allowance. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving wall lengths. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving material ordering. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving mortar mixing. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving wastage allowance. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving floor areas. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving wall lengths. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving material ordering. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving mortar mixing. Show formula, substitution, units and interpretation.

3. Percentages

Wastage, discount, variation and increase/decrease.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\mathrm{Ratio} = a:b$$

Worked method

Example: Perimeter. Find skirting length for a rectangular room of 4.5 m by 3.2 m.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports floor areas. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving material ordering. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving mortar mixing. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving wastage allowance. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving floor areas. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving wall lengths. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving material ordering. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving mortar mixing. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving wastage allowance. Show formula, substitution, units and interpretation.

4. Measurement

Length, area, volume and unit conversion.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$P=2(l+w)$$

Worked method

Example: Mortar ratio. A 1:4 mortar mix requires 50 kg cement. Estimate sand required.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports wall lengths. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving mortar mixing. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving wastage allowance. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving floor areas. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving wall lengths. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving material ordering. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving mortar mixing. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving wastage allowance. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving floor areas. Show formula, substitution, units and interpretation.

5. Building applications

Material checks and site arithmetic.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$A=lw$$

Worked method

Example: Wastage. Add 8% wastage to 120 m² of floor tiles.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports material ordering. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving wastage allowance. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving floor areas. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving wall lengths. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving material ordering. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving mortar mixing. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving wastage allowance. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving floor areas. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving wall lengths. Show formula, substitution, units and interpretation.

Mixed Revision Questions

1. A building technology problem involves wastage allowance. Choose a suitable mathematical method, solve the problem, and explain the result.
2. A building technology problem involves floor areas. Choose a suitable mathematical method, solve the problem, and explain the result.
3. A building technology problem involves wall lengths. Choose a suitable mathematical method, solve the problem, and explain the result.
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10. A building technology problem involves mortar mixing. Choose a suitable mathematical method, solve the problem, and explain the result.
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12. A building technology problem involves floor areas. Choose a suitable mathematical method, solve the problem, and explain the result.
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15. A building technology problem involves mortar mixing. Choose a suitable mathematical method, solve the problem, and explain the result.
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17. A building technology problem involves floor areas. Choose a suitable mathematical method, solve the problem, and explain the result.
18. A building technology problem involves wall lengths. Choose a suitable mathematical method, solve the problem, and explain the result.
19. A building technology problem involves material ordering. Choose a suitable mathematical method, solve the problem, and explain the result.
20. A building technology problem involves mortar mixing. Choose a suitable mathematical method, solve the problem, and explain the result.
21. A building technology problem involves wastage allowance. Choose a suitable mathematical method, solve the problem, and explain the result.

22. A building technology problem involves floor areas. Choose a suitable mathematical method, solve the problem, and explain the result.
23. A building technology problem involves wall lengths. Choose a suitable mathematical method, solve the problem, and explain the result.
24. A building technology problem involves material ordering. Choose a suitable mathematical method, solve the problem, and explain the result.
25. A building technology problem involves mortar mixing. Choose a suitable mathematical method, solve the problem, and explain the result.
26. A building technology problem involves wastage allowance. Choose a suitable mathematical method, solve the problem, and explain the result.
27. A building technology problem involves floor areas. Choose a suitable mathematical method, solve the problem, and explain the result.
28. A building technology problem involves wall lengths. Choose a suitable mathematical method, solve the problem, and explain the result.
29. A building technology problem involves material ordering. Choose a suitable mathematical method, solve the problem, and explain the result.
30. A building technology problem involves mortar mixing. Choose a suitable mathematical method, solve the problem, and explain the result.

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.
2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.
3. Uploaded KNEC solved mathematics materials for examination-style orientation.
4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.